



RUMLEM: Romansh Lemmatizer

University of Zurich – Department of Computational Linguistics | Dominic P. Fischer – Zachary Hopton – Jannis Vamvas

DEMO

Click to analyse unknown words
by using common patterns

Input Text

La vulp gronda
era puspè ina
giada fomentada

Analyze

Detected Idioms

Rumantsch Grischun

Sursilvan

Sutsilvan

Surmiran

Puter

Vallader

Percentage of words found

10 20 30 40 50 60 70 80 90

Analysis of Words

Token	Lemma	German Translations	Morphological Analysis
La	la	die, ihr, sie, ...	—
vulp	vulp	Fähe, Fuchs, Pfiffikus, ...	PoS= N ; Gender= FEM ; Number= SG
gronda	gronda (?) grond	gross, dick,	PoS= ADJ ; Gender= FEM ; Number= SG
era	era esser	Beet, Rabatte, ... sein, sich befinden, ...	PoS= N ; Gender= FEM ; Number= SG PoS= V ; Mood= IND ; Tense= PST ; Person= 1/3 ; Number= SG ; Aspect= IPFV
puspè	puspè	noch einmal, wieder, ...	—
ina	ina	eine(r, -s), irgendeine	—
giada	giada	Mal	PoS= N ; Gender= FEM ; Number= SG
fomentada	fomentà fomentar	hungrig aushungern	PoS= ADJ ; Gender= FEM ; Number= SG PoS= V ; VerbForm= PTCP ; Tense= PST ; Gender= FEM ; Number= SG

EVALUATION

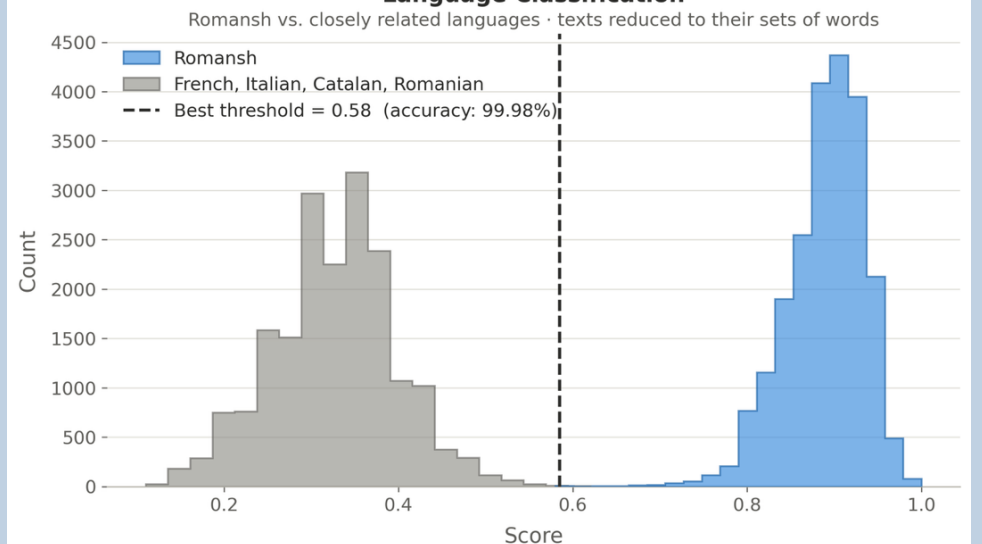
	Variety Classifi- cation (%)	Coverage (%)	Extended Coverage (%)
Rumantsch Grischun	100	79	86
Sursilvan	86	84	86
Sutsilvan	100	77	84
Surmiran	95	82	88
Puter	98	84	89
Vallader	92	80	84

Variety Classification: % of text samples correctly classified into their variety

(Extended) Coverage: % of lemmatizable words w/ and w/o applying learned patterns to unk. words

Language Classification: Plotting of highest Romansh variety score for each text shows separability
These results are based on 30'000+ samples of text, ranging from a few words to as many as 2000.

Language Classification



About RUMLEM

RUMLEM is a dictionary-based, variety-aware lemmatizer for the five main Romansh idioms as well as the supra-regional standard variety Rumantsch Grischun. The data for Rumantsch Grischun, Surmiran, Sursilvan and Sutsilvan are openly licensed (© Lia Rumantscha 1980–2025). The Vallader and Puter dictionaries were kindly provided by Uniun dals Grischs for use solely as part of this lemmatizer (© Uniun dals Grischs. All rights reserved).

We thank the Swiss Federal Office of Culture (Bundesamt für Kultur BAK) for their support in the realisation of this project.

Code: github.com/ZurichNLP/rumlem

Demo: huggingface.co/spaces/ZurichNLP/rumlem



Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Eidgenössisches Departement des Innern EDI
Bundesamt für Kultur BAK